
Quantitative Approach to Diversity and Decline in Late Palaeozoic Trilobites

Rudy Lerosey-Aubril and Raimund Feist

Abstract

Quantitative data reveal complex evolution in late trilobite diversity. In the Middle to early Late Devonian, a series of extinction events led to dramatic taxonomic impoverishment of the trilobites. In the Famennian, when only two orders remained, originations began to compensate for the still high extinction rates, marking the start of a remarkable diversification. Although interrupted by the major Hangenberg turnover, the general diversification trend accelerated in the Tournaisian, whereas extinctions became modest. Originations diminished markedly during the Viséan and Serphukovian, causing this diversity to decrease to the level observed in the Frasnian. It was never much higher thereafter, despite massive restructuring of the trilobite communities in the early Pennsylvannian; this allowed for progressive domination of the ditomopygines. After another decline in the Kasimovian, a period of stasis occurred with very low diversity levels and almost no renewal. The last burst of diversification occurred in the Wordian, but ceased rapidly in the Capitanian when degradation of environmental conditions began to inhibit originations. Thus, extinction of the Trilobita at the end of the Permian resulted from disappearance of merely a handful of genera.

Keywords

Trilobites • Devonian-Permian extinctions • Quantitative analysis • Early Devonian diversity peak • Mid-Devonian–Frasnian decrease • Famennian–Tournaisian diversity acceleration • Viséan–Serpukhovian marked diversity decrease • Last diversification Wordian • Decreasing diversity Capitanian to end-Permian total extinction

Introduction

The extraordinary evolutionary success of the arthropods is incontestable. Present in virtually all habitats, frequently in great abundance, arthropods represent by far the most-diversified group of metazoans on Earth. Their abundance, taxonomic diversity, and omnipresence all demonstrate that these organisms

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have a unique capacity for adaptation. However, about 250 million years ago, a particularly diversified and emblematic group of Palaeozoic arthropods became extinct, the trilobites.

In contrast to most representatives of the phylum that appeared during the Cambrian explosion, trilobites diversified rapidly to become a dominant element in marine benthic communities. Throughout the Early to early Middle Palaeozoic, their diversity fluctuated with an apogee recorded in the Middle Cambrian and minimum values during the Silurian (Adrain 2008). Another peak of trilobite diversity occurred during the Early Devonian with values similar to those of the Late Cambrian. Probably because the diversity and abundance of these organisms decreased dramatically thereafter, much less attention has been accorded to their late evolutionary history, although this history offers a unique opportunity to study an exceptional event in the history of life: the complete extermination of a major group of arthropods. It has long been known that a series of biological events during the Middle to Late Devonian caused a dramatic reduction in trilobite diversity at high taxonomic levels so that the class came to be represented by the single order, Proetida, throughout the Late Palaeozoic. However, to date, neither the various steps that led to the almost inevitable extinction of the Trilobita during the Permo-Triassic mass extinction nor the impacts of the major climatic, eustatic, and tectonic events that characterized the late Middle to Late Palaeozoic have been described in detail.

In this chapter, we take advantage of the recent revision of the stratigraphic framework of the Devonian (Bultynck 2000), Carboniferous (Heckel and Clayton 2006), and Permian (Owens 2003) to develop a precise quantitative description of the evolution of trilobite diversity from the Middle Devonian to the terminal Permian, based on a recently compiled taxonomic database. Because we consider the extinction of the trilobites to be a historical event that must be restored to its historical context, we then attempt to interpret the observed diversity patterns with special reference to global biotic, climatic, eustatic, and tectonic events.

Methodology

To allow a quantitative approach to the evolution of trilobite diversity from the Middle Devonian to the terminal Permian, a taxonomic database has been

created that re-groups all the genera and subgenera known from this period. The data for the Devonian were initially extracted from earlier works (essentially Briggs et al. 1988; Feist 1991; and Chlupáč 1994) and then updated and completed with more recent data. Additions to the database have allowed us to consider almost twice the number of taxa considered by Chlupáč (1994). The database for the Carboniferous and Permian is entirely new and was compiled from primary sources. Throughout the compilation, no distinction was made between genera and subgenera; all are cited as 'genera' below. Occurrences for which the assignment and/or the age of the taxa were questionable were not used to describe the diversity patterns presented here, although curves representing the evolution of raw generic diversity seem to be comparable whether or not these occurrences are considered. The time intervals considered are the stages throughout the study. All in all, our database re-groups 425 taxa in total: Mid-Late Devonian 226; Carboniferous 177; Permian 35), this stratigraphic distribution has been reviewed and updated with reference to the most recent stratigraphic framework proposed by the International Commission of Stratigraphy (Ogg et al. 2008).

To propose an accurate description and analysis of the diversity patterns of the Middle Devonian to Permian trilobites, we have used various metrics and evaluated the concordance of the results achieved. Four series of measures were used: raw taxonomic richness, standing diversity, normalized diversity (i.e. raw diversity divided by the duration of time interval), and origination/extinction/turnover rates. For standing diversity (SDv1, SDv2), origination rate (OR1, OR2, SOR), extinction rate (ER1, ER2, SER), and turnover rate (TuR1, TuR2, STR), the metrics defined by Harper (1996; see his Table 1 for definitions) have been used.

Description

Diversity

In the later part of the Palaeozoic, trilobite diversity reached its maximum in the Early Devonian when trilobites were nearly as globally diverse as during the Late Cambrian (Adrain 2008). Following this period of abundance, culminating in the terminal Emsian, the Middle Devonian begins with still high raw ordinal and familial diversities (Fig. 1a). However, thereafter

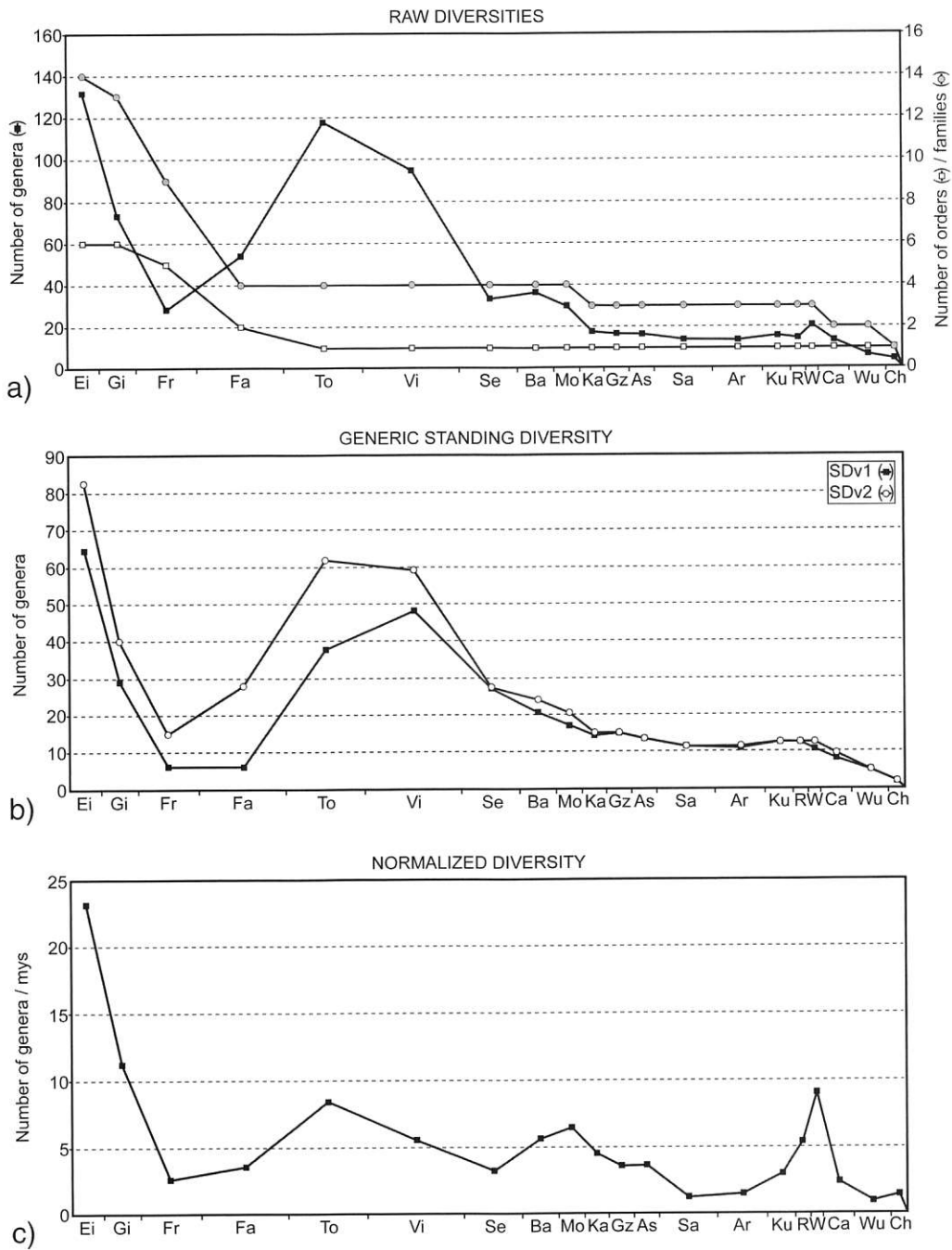


Fig. 1 Evolution of trilobite diversity from the Eifelian (Middle Devonian) to the Changhsingian (Lopingian). **(a)** Ordinal, familial, and generic raw diversities. **(b)** Generic standing diversity. SDv1 and SDv2 are the two different metrics defined by Harper (1996). **(c)** Raw generic diversity normalized to the duration of the time interval in millions of years. Except for the Roadian (R) and the Wordian (W), the names of the

stratigraphic stages are abbreviated to their first two letters: Eifelian (Ei), Givetian (Gi), Frasnian (Fr), Famennian (Fa), Tournaisian (To), Viséan (Vi), Serpukhovian (Se), Bashkirian (Ba), Moscovian (Mo), Kasimovian (Ka), Gzhelian (Gz), Asselian (As), Sakmarian (Sa), Artinskian (Ar), Kungurian (Ku), Capitanian (Ca), Wuchiapingian (Wu), Changhsingian (Ch)

and up to the Devonian–Carboniferous (D/C) boundary, there was a pronounced reduction in diversity of the higher taxonomic categories when extinctions (5 orders, 11 families, 30 subfamilies) greatly outnumbered originations (0 orders, 1 family, 9 subfamilies). Thus, at the onset of the Upper Palaeozoic, trilobites were represented by only one order (Proetida) and four families. This low diversity at high taxonomic levels lasted until the extinction of the group during the end-Permian mass extinction, with all but one family disappearing after the Wordian.

The evolution of raw generic diversity is more complex (Fig. 1a), with two phases discernible in the Middle through Late Devonian and four in post-Devonian times. The first phase, from the Eifelian to the Frasnian, was characterized by a substantial loss of diversity. During the Famennian, their diversity increased again, a trend that continued through the Tournaisian when values reach more than 85% of those observed in the Eifelian. This period was followed by an initially moderate decline during the Viséan, becoming more severe during the Serpukhovian when the number of genera was similar to that in the Frasnian. The early Pennsylvanian (Bashkirian) was characterized by a modest rebound, followed immediately by another fall during the Moscovian and Kasimovian. Raw generic diversity then remained invariably low until the beginning of the Guadalupian ('middle Permian'). A final limited increase occurred during the Wordian, but once again, it was followed immediately by a decline that persisted until the Permian–Triassic boundary.

The pattern of evolution of the standing generic diversity of the trilobites from the Middle Devonian to the end of the Permian differs markedly when stratigraphic singletons (i.e. taxa confined to a single time interval) are considered. When they are considered (SDv2), the evolution of the standing generic diversity is remarkably similar to that of the raw generic diversity (Fig. 1b), although the values are invariably lower. The main differences include relatively higher levels in the Viséan and Serpukhovian and a certain flattening of the curve thereafter; it lacks the clear diversity rebounds that occurred in the Bashkirian and Wordian. When the stratigraphic singletons are ignored (SDv1), the standing and raw diversities differ to a greater extent (Fig. 1b). The post-Viséan part of the curve is even more flattened; the values

reached between the Eifelian and Viséan are significantly lower, especially during the Tournaisian and Famennian.

Normalizing measures of raw generic diversity by dividing them by the duration of the time intervals (in millions of years) markedly modifies the pattern of evolution of generic diversity (Fig. 1c). From the Eifelian to the Frasnian there was a sharp continuous decline in the normalized generic diversity; this is mirrored in the evolution of the raw and standing diversities during the same period (Fig. 1a, b). However, during the Famennian, there was no noticeable increase in the normalized diversity or even during the Tournaisian when its rise remained modest compared with rises in the raw and standing diversities. Its peak was only one-third the value of the Eifelian peak. The normalized diversity then decreased evenly to a minimum of 3.14 genera per million years in the Serpukhovian. The Bashkirian rebound in the normalized diversity was more pronounced; it even continued during the Moscovian; the decline commenced only during the Kasimovian. Unlike the raw and standing diversities, which plateaued from the Kasimovian to the Roadian (Fig. 1a, b), the normalized generic diversity dropped sharply between the Asselian and Sakmarian. This decline led to extremely low values; this situation persisted until the Kungurian and the beginning of the last period of increasing diversity. This period culminated during the Wordian when the diversity per million years was the highest in all post-Devonian times, confirming that diversification did occur during the Middle Permian; this is perhaps suggested by the pattern of evolution (Fig. 1a). However, a severe drop immediately followed this Wordian diversification lead to an extremely low normalized diversity for the whole Lopingian ('Late Permian').

Origination, Extinction, and Turnover Rates

Once again, considering (OR1, ER1, TuR1) or ignoring (OR2, ER2, TuR2, SOR, SER, STR) stratigraphic singletons substantially modifies the evolutionary patterns of origination, extinction, and turnover rates. When singletons are considered, the origination rate (OR1; Fig. 2a) had a moderately low value (0.4) in the Middle Devonian, increasing strongly throughout the Late Devonian to reach 0.96 during the Famennian.

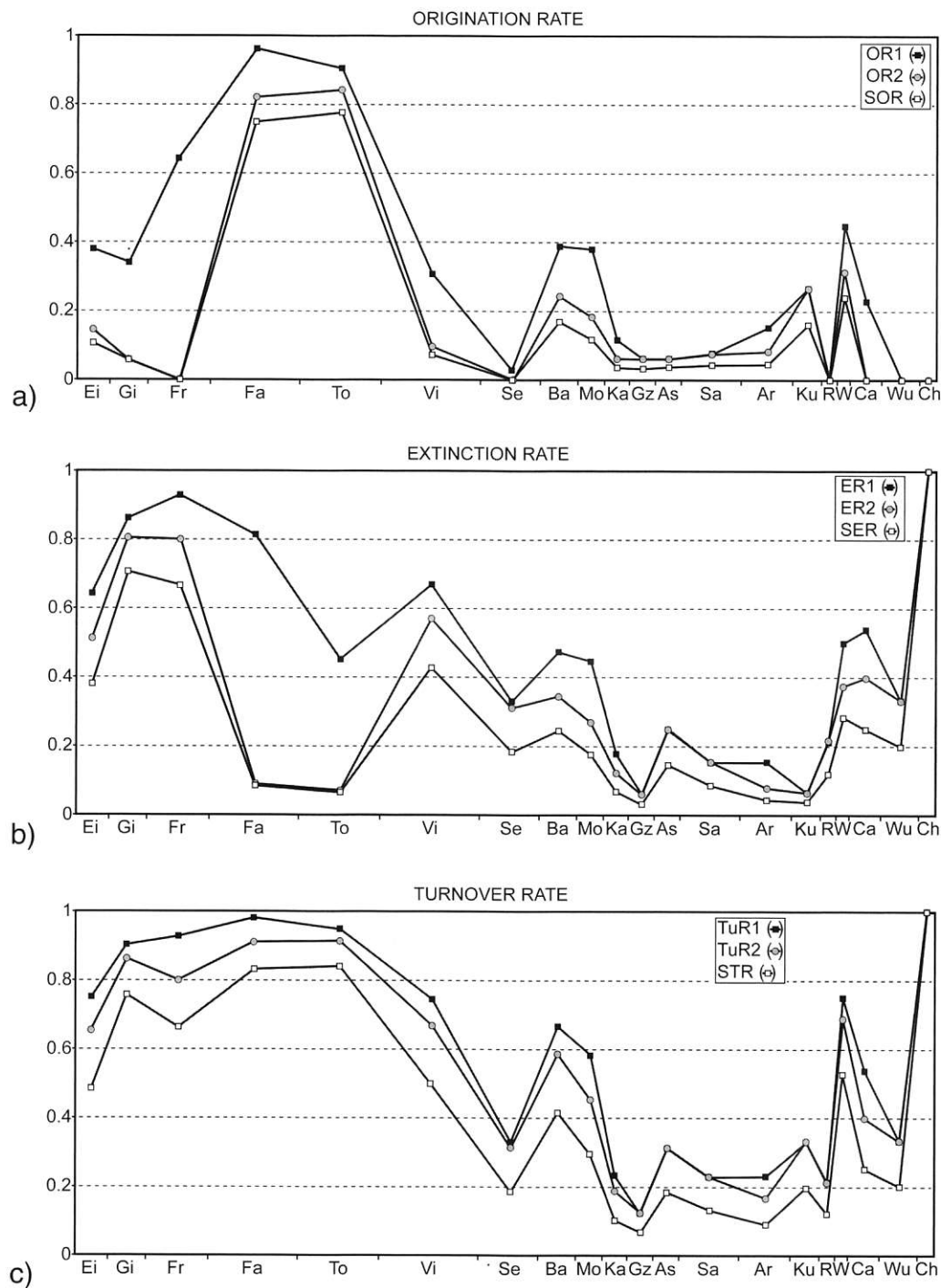


Fig. 2 Evolution of the origination (a), extinction (b), and turnover (c) rates of trilobites from the Eifelian (Middle Devonian) to the Changhsingian (Lopingian). The various metrics used to describe the origination rate (OR1, OR2, SOR),

extinction rate (ER1, ER2, SER), and turnover rate (TuR1, TuR2, STR) have been defined by Harper (1996: Table 1). See the caption of Fig. 1 for abbreviations designating the stratigraphic stages

Roughly unchanged during the Tournaisian, the origination rate fell drastically thereafter, approaching zero by the end of the Mississippian. Markedly higher origination rates (0.4) occurred during the Bashkirian–Moscovian interval, but only temporarily because a strong fall occurred during the Kasimovian marking the onset of a period of low OR1 (~ 0.1) that lasted until the Sakmarian. An increase then occurred from the Artinskian to the Wordian (OR1 = 0.45), but it was interrupted by a brutal drop to zero during the Roadian. This last episode of elevated origination rates was immediately followed by a decline with no origination occurring during the Lopingian. Ignoring the stratigraphic singletons leads to very similar patterns of evolution of the origination rate (OR2) and the standing origination rate (SOR), the main difference being slightly lower values for SOR (Fig. 2a). These two patterns were also similar to that observed with OR1 for most of the post-Devonian period, despite a certain flattening of the curves. However, OR2 and SOR were much lower than OR1 during the Middle and Late Devonian, particularly during the Frasnian when they declined to zero, whereas OR1 was close to 0.65.

When singletons are considered (ER1), the extinction rate was high throughout the Middle and Late Devonian, initially increasing from 0.65 (Eifelian) to more than 0.9 (Frasnian) before decreasing slightly to 0.8 during the Famennian (Fig. 2b). A marked decrease occurred during the Tournaisian (~ 0.45) followed by an increase during the Viséan (~ 0.7) and another drop during the Serpukhovian (~ 0.35). After a plateau around 0.45 during the early Pennsylvanian, ER1 fell to below 0.2 during the Kasimovian; this marked the onset of a period of low but fluctuating values (from 0.05 to 0.25) lasting until the Roadian. From the Roadian to the end of the Permian, ER increased significantly despite a marked drop during the Wuchiapingian. When singletons are ignored (ER2 and SER), the extinctions once again display extremely similar evolutionary patterns, differing significantly from those of ER1 only during the Famennian and Tournaisian (Fig. 2b). ER2 and SER were extremely low (~ 0.1) during this time interval whereas ER1 was between 0.8 (Famennian) and 0.45 (Tournaisian).

The differences in the calculated origination and extinction rates that occur when stratigraphic singletons are or are not considered are somewhat attenuated when the turnover rates are calculated. Indeed, whether singletons are considered (TuR1) or not (TuR2, STR),

the evolutionary patterns of the turnover rates are fundamentally the same (Fig. 2c), except that the standing turnover rate (STR) has constantly lower values than the simple turnover rates (TuR1, TuR2). From the Eifelian to the Tournaisian the turnover rates were all rather high, even when TuR2 and STR showed a period of lower values centred on the Frasnian when TuR1 remained above 0.9. The turnover rates decreased during the Viséan and Serpukhovian, reaching low values during the latter stage. In contrast, the Bashkirian–Moscovian period represents one of the three post-Mississippian peaks in turnover rates, together with those in the Wordian and Changhsingian. It was followed by an interval of rather low (~ 0.35) to very low (~ 0.1) turnover rates lasting until the Roadian. Similarly, the Wordian and Changhsingian peaks in turnover rates flank a period of lower but still significant turnover rates.

Composition

The evolution of trilobite composition from Mid to Late Palaeozoic times is shown in Fig. 3. With 6 orders and 14 families, the trilobites still represented a diversified group of marine benthic invertebrates during the Eifelian (Fig. 1a). The orders Phacopida and Proetida already dominated the trilobite communities, representing 34 and 44% of the raw generic diversity, respectively (Fig. 3). The contribution of the Phacopida to trilobite diversity increased slightly until the Frasnian with the Acastoidea outnumbering the Phacopoidea during the Middle Devonian. The Proetida, dominated by proetoids and in particular the dechenellines until the Givetian, regressed moderately. However, all in all, these two orders constituted more than 75% of the generic diversity throughout the Middle and early Late Devonian. Similarly, the contribution of the Corynexochida fluctuated slightly between 5.5 and 10% during this period. In contrast, contributions of the Odontopleurida and Harpetida increased moderately during the same period, whereas that of the Lichida decreased. This latter order also became extinct during the Givetian. Predominance of the Proetida and the Phacopida was also expressed at the familial level (Fig. 4). Whereas the other orders were represented by a single family from the Eifelian to the Frasnian, four proetid families and two to six phacopid families occurred during this period

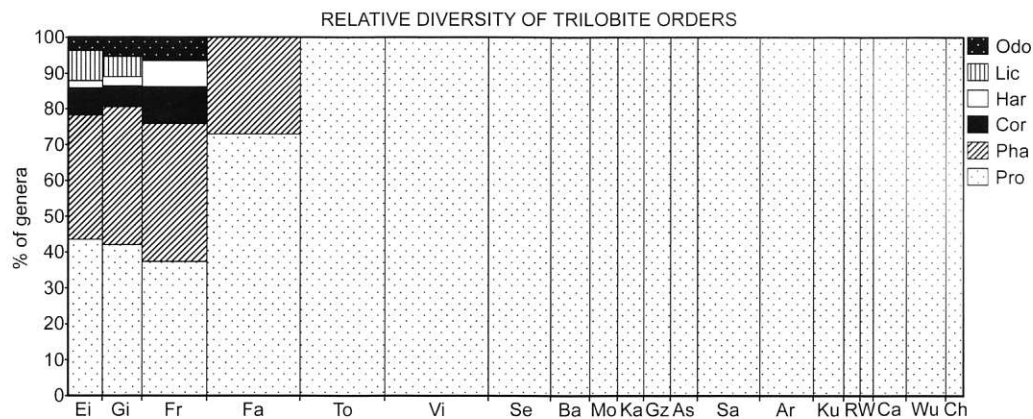


Fig. 3 Relative diversities of the trilobite orders from the Eifelian (Middle Devonian) to the Changhsingian (Lopingian). Abbreviations of orders: Corynexochida (Cor), Harpetida (Har), Lichida (Lic), Odontopleurida (Odo), Phacopida (Pha), Proetida (Pro). See the caption of Fig. 1 for the abbreviations designating the stratigraphic stages

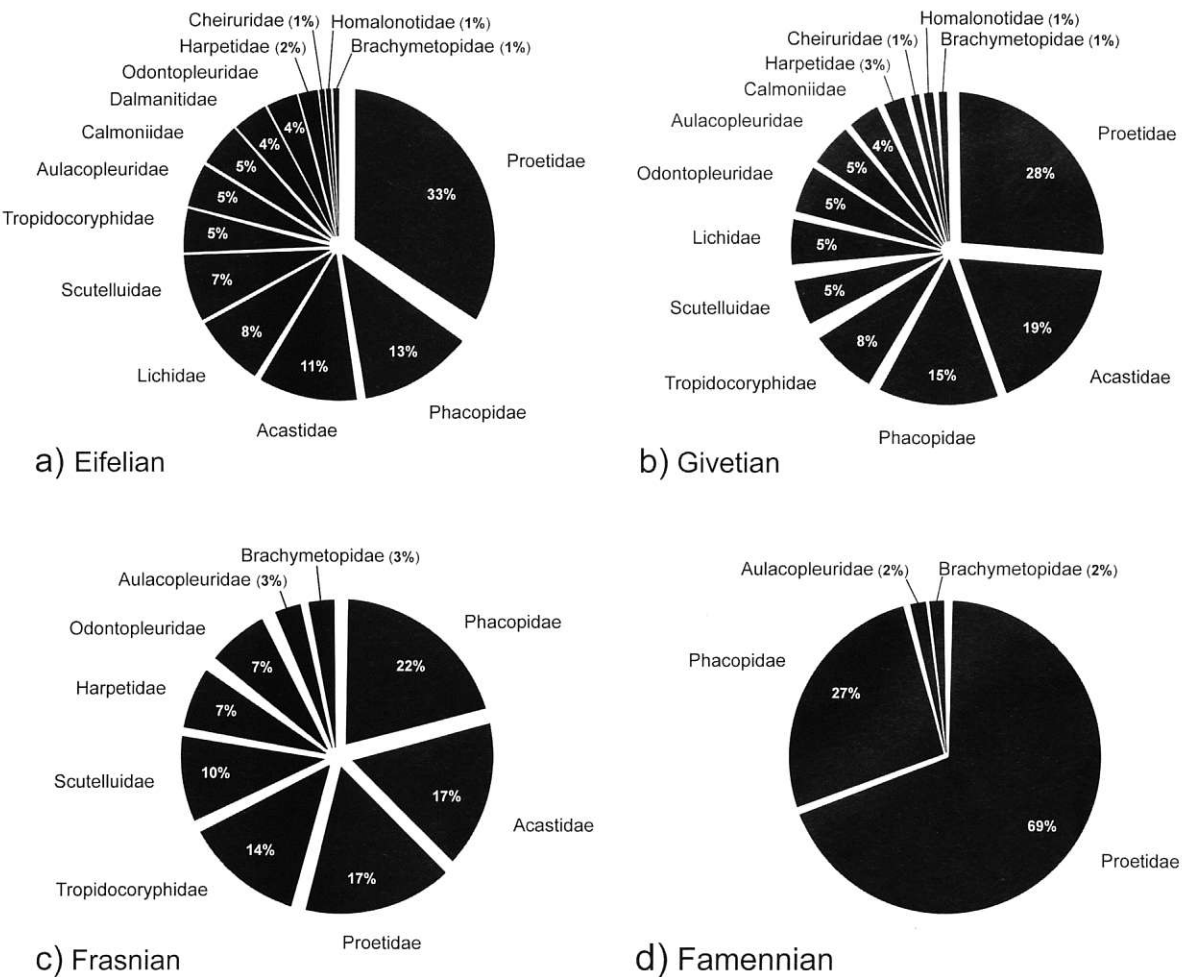


Fig. 4 Relative diversities of trilobite families during the Eifelian (a), Givetian (b), Frasnian (c), and Famennian (d)

(Fig. 4a–c). In this regard, the Frasnian is noted for phacopids because it was associated with a significant fall in raw familial diversity. The Phacopidae and Proetidae were the only families that invariably displayed high generic diversity until the Frasnian (Fig. 4a–c).

The Famennian was characterized by a severe decline in trilobite diversity at high taxonomic levels; only two orders and four families were present during this period (Figs. 1a, 4d). End-Frasnian extinctions greatly benefited the Proetida; they then contributed more than 70% of the total generic diversity (Figs. 3, 4d). The Proetidae diversified strongly to constitute nearly 85% of the raw generic diversity during the Eifelian.

The Tournaisian was another period when the trilobite composition was profoundly modified. Not only was the Proetida the sole order present at the beginning of the Carboniferous (Fig. 3), but trilobite communities were totally dominated by a new family, the Phillipsiidae; it represented almost 90% of the generic diversity. This domination lasted throughout the Late Palaeozoic, its contributions to generic diversity fluctuating between 70% (Artinskian) and 100% (Changhsingian). The respective contributions to trilobite diversity of the three families directly inherited from the Devonian remained roughly unchanged until their extinctions: about 2% for the Aulacopleuridae, 6% for the Proetidae, and 11% for the Brachymetopidae. None of these families lasted with certainty until the terminal Permian. The aulacopleurids, proetids, and brachymetopids became extinct during the Moscovian, Wordian, and Wuchiapingian, respectively.

If the trilobite composition fluctuated negligibly at family level during the Carboniferous and Permian, this was not the case at subfamily level. The Phillipsiidae dominated the trilobite communities tremendously throughout this period, so evolution of trilobite composition at that time can largely be explained by changes in the relative diversity of the phillipsiid subfamilies (Fig. 5). Phillipsiid trilobites were particularly diversified in the early Mississippian with a maximum of nine subfamilies. Of these, the Archegoniinae were the most abundant, representing about 30% of the raw generic diversity of the phillipsiids until the Bashkirian. Their relative diversity then declined strongly during the Moscovian; only a single genus has been described

from younger (Wordian) strata. The Cystispiniinae also were a rather important group in the Tournaisian, but its relative diversity decreased progressively thereafter until its extinction at the end of the Mississippian. Weaniine trilobites were rather diverse at the beginning of the Carboniferous; they remained so until their extinction at the end of the Guadalupian, with a peak of relative diversity during the Artinskian. However, the Linguaphillipsiinae and Phillipsiinae were minor contributors to the Tournaisian diversity of the trilobites; both groups became extinct before the end of the Mississippian. The stratigraphic range of the Griffithidae extended into the Moscovian, but their relative diversity remained rather modest throughout this period, exceeding 10% only during the Serpukhovian. Lastly, the Ditomopyginae, Bollandiinae, and Cummingellinae were the only subfamilies to survive until the terminal Permian. Among them, the Bollandiinae was a significant contributor to phillipsiid diversity only during the Lopingian when trilobite raw diversity became very low. Likewise, cummingelline trilobites appeared relatively diversified during this period; they also represented about 20% of the phillipsiid diversity during the Kasimovian and early Cisuralian. However, of all the phillipsiid subfamilies, the most important was definitely the Ditomopyginae. The contribution of this group to phillipsiid generic diversity increased progressively during the Mississippian, reaching 25% during the Bashkirian. Thereafter it was by far the most diversified trilobite subfamily until the extinction of the class, rarely representing less than 50% of all phillipsiid diversity.

Analysis

The Middle Devonian to Early Late Devonian Period

Whatever metrics are considered, this first phase of the evolution of trilobite diversity was characterized by a substantial decline at all taxonomic ranks (Fig. 1). This decline resulted mainly from the particularly high extinction rates, whereas the origination rates oscillated simultaneously from 0.35 to 0.65 at best (Fig. 2). However, the compositions of the trilobite communities changed little at the ordinal level; the Phacopida and Proetida were

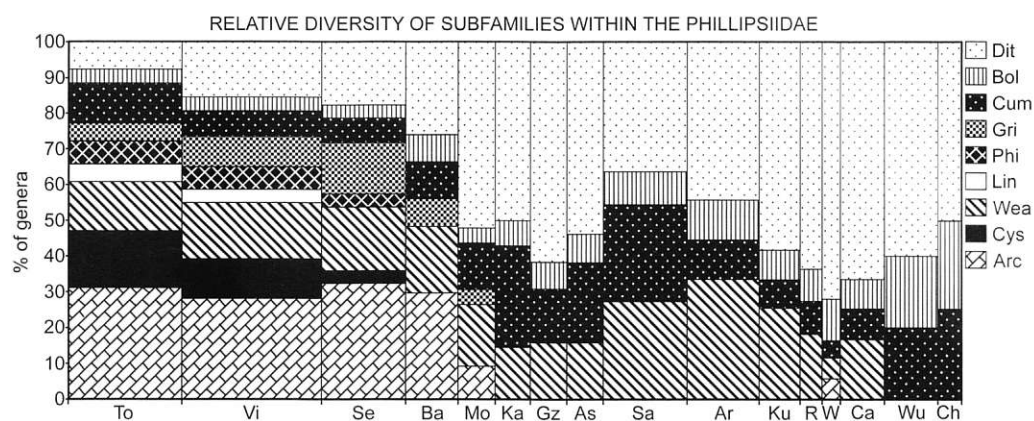


Fig. 5 Relative diversities of subfamilies within the trilobite family Phillipsiidae from the Tournaisian (Early Carboniferous) to the Changhsingian (Lopingian). Abbreviations of subfamilies: Archegoninae (Arc), Bollandiinae (Bol), Cummingellinae

(Cum), Cystispinae (Cys), Ditomopyginae (Dit), Griffithidinae (Gri), Linguaphillipsiinae (Lin), Phillipsiinae (Phi), Weaniinae (Wea). See caption of Fig. 1 for abbreviations designating stratigraphic stages

enormously dominant throughout this time interval (Fig. 3).

Figure 1 shows the evolution of trilobite diversity as a continuous decline from the Eifelian to the Frasnian but, in fact, this evolution was stepwise with more or less drastic diversity changes alternating with intervals of stasis. Severe environmental perturbations repeatedly affected the marine environments during this period, generating severe restructuring within marine ecosystems on a global scale (Walliser 1996). Because most of these bioevents did not strictly coincide with stratigraphic stage boundaries and because several of these events occurred within a single stage, a correct assessment of the evolutionary dynamics of trilobite diversity during this first phase requires examination of shorter time intervals. Eight biotic events, with a mean extinction rate of 30 genera per event, can be recognized from the Eifelian to the end-Frasnian (Fig. 6). The terminal Emsian and basalmost Eifelian was a time of apogee for taxa such as the Asteropyginae, Calmoniidae, Odontopleuridae, Phacopinae, Synphoriinae, and Scutelluidae (Chlupáč 1994). The series of environmental perturbations that affected their habitats during the Middle and early Late Devonian reduced their diversity to a such extent that three of the five orders present in the Frasnian were represented by a single genus immediately before the Upper Kellwasser Event.

Both Eifelian events, the Choteč Event and the Kačák Event, might have resulted from global eustatic rises (Johnson et al. 1985) coupled to establishment

of oxygen-depleted conditions in benthic habitats. As a consequence, the degree of diversification remained roughly stable compared with that in Early Devonian times, though it was rather low in groups such as the Homalonotidae, Cheiruridae, and Brachymetopidae. However, adaptive innovations occurred in the Proetidae, Acastidae, and Lichidae; the last included double the number of genera in the Eifelian compared to what it had at the end of the Early Devonian.

With the terminal mid-Givetian Taghanic Onlap, the mean sea-level reached the highest stands of all Mid to Late Palaeozoic times (Haq and Schutter 2008), leading to a worldwide expansion of the pelagic environment and to the demise of a great variety of near-shore neritic and peri-reefal habitats. Trilobites diversified particularly on the inner platform, especially in near-shore to coastal environments; these exhibited abrupt lateral variations in their physico-chemical conditions. The abolition of these habitats might largely explain the severe extinctions associated with the Taghanic Event; this was connected with a considerable reduction in generic and specific trilobite abundances (Fig. 6) and also to the disappearance of several emblematic families such as the Calmoniidae, Cheiruridae, and Lichidae, together with major proetoid (Eremiproetinae, Dechenellinae) and aulacopleurid (Aulacopleurinae, Cyphaspidinae) subfamilies. In North America, the Homalonotidae, Dechenellinae, and certain Asteropyginae (*Greenops*) also become extinct, but

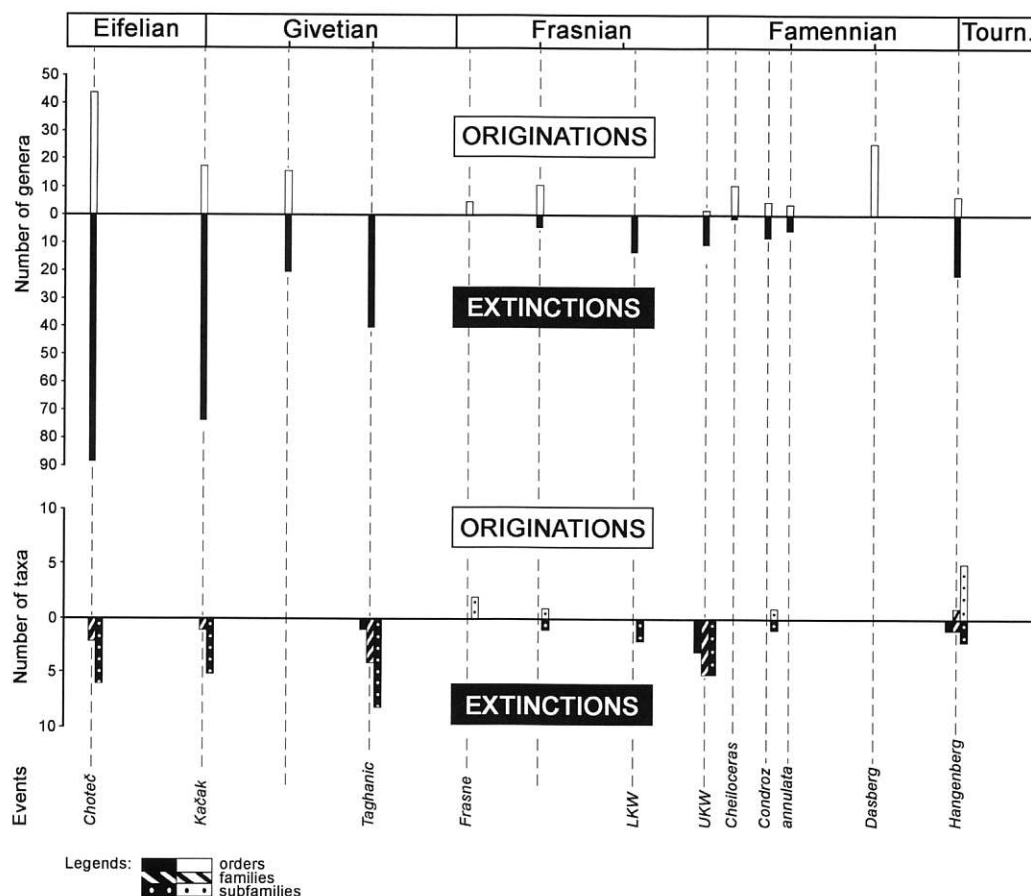


Fig. 6 Number of originations and extinctions of genera (*upper part*) and subfamilies, families, and orders (*lower part*) during the Middle and Late Devonian biotic events. Tournaisian is abbreviated to Tourn.

probably a little later than the middle *varcus* Zone, although still within the Givetian. For unknown reasons, the Taghanic Event apparently affected trilobite communities, in particular those of the European/North African realm, earlier than other benthic biotas such as corals/stromatoporoids (within the Late *varcus* Zone) and brachiopods (between the Early and Late *hermanni-cristatus* zones) (Walliser 1996).

A few taxa survived this world-wide ocean deepening and successfully adapted to the deeper and more homogeneous environmental conditions in the stylioline–cephalopod realm of the late Givetian. In contrast to the pre-Taghanic Early and Mid Devonian trilobite communities – which were rather diverse, locally extremely abundant, and typically distributed in geographically restricted biofacies – the surviving faunas exhibited a pronounced reduction in generic and specific diversity concomitant with an increase in

cosmopolitanism. Therefore, all the families/subfamilies that occurred in the stylioline–cephalopod realm were composed of single genera and radiated into the Late Devonian. In summary, with extremely low generic origination (zero) and extinction rates, the post-Taghanic Givetian times marked the onset of a period of stability with regard to trilobite diversity; this persisted during most of the Frasnian (Fig. 6).

At the beginning of the Late Devonian, the Frasnian or *Manticoceras* Event was not marked by notable extinctions of trilobites (Fig. 6), whereas it had an important impact on organisms such as ammonoids (extinction of most pharciceratids, appearance of *Manticoceras*), brachiopods, corals, and stromatoporoids (Walliser 1996). Among the 27 known trilobite genera in the Frasnian, only 5 originated in the early Frasnian; 11 others appeared during the mid-Frasnian. Both generic and specific diversities remained relatively

low during the 10 Ma or so of the global sea-level high stand that preceded the terminal Frasnian extinction events (Haq and Schutter 2008). This stability in trilobite diversity may have mirrored the homogenization of the environmental conditions on the expanding outer platforms, resulting from the loss of oceanic barriers (e.g. the remnants of the Rheic ocean) caused by rapprochement of the epicontinental margins. In these outer shelf environments, the biotic evolutionary responses to extrinsic constraints generated a paedomorphic trend towards eye reduction in many lineages, as in the proetoid *Pteroparia* and *Palpebralia* (Feist 1991, 1995). Although no longer present on the African and American continents, large-eyed asteropygines experienced a period of pronounced specific diversification in the early and mid-Frasnian neritic brachiopod-rich organo-clastic environments extending from Europe to Southern Asia (Morzadec 1992). Interestingly, *Bradycryphaeus*, the only asteropygine to reach the base of the Upper Kellwasser Extinction level, lived within the off-shore cephalopod milieu.

The terminal Frasnian Kellwasser Extinction Event particularly affected trilobites adapted to life in deep epibenthic habitats. Of the five orders present during the Frasnian (Corynexochida, Harpetida, Odontopleurida, Phacopida, and Proetida), only the Phacopida and Proetida persisted into the Famennian. In European and North African localities, the disappearance of the trilobite communities coincided with a period of eustatic instability; it triggered a sudden spread of oxygen-depleted depositional conditions in the deeper outer-shelf areas. Such hypoxic deposits, the so-called Lower and Upper Kellwasser horizons, occurred at the end of conodont Zone 12 and during Zone 13c, respectively (Girard et al. 2005). Precise correlations of the last occurrences, which reached the base of the Upper Kellwasser extinction level, have been obtained for several sites in Europe (Feist and Schindler 1994) and North Africa (Feist 2002); they are uniformly fine-grained, mostly calcilititic, cephalopod- and dactyloconarid-rich deposits representative of off-shore depositional environments at or beyond the limit of light penetration. By comparison, late Frasnian trilobites from near-shore deposits before and contemporaneous with the Kellwasser Extinction levels remain poorly known. At present, no data exist for the Americas or Asia, for example. In this context, the faunas from the Canning Basin (NW Australia) are essential because they occur in precisely dated

sections where well-oxygenated conditions prevailed throughout a continuous Frasnian–Famennian transition (George and Chow 2002). These comprise carbonate ramp and basinal deposits with biogenic debris flows; these originated from immediately adjacent shallow reefal and peri-reefal areas. In contrast to the European and North African sections, the continuous Frasnian–Famennian transitional beds are devoid of any Kellwasser-type hypoxic or anoxic intercalations. The terminal Frasnian extinction event can nevertheless be recognized by a biotic loss contemporaneous with and of similar intensity to those observed elsewhere (McNamara and Feist 2008). The late Frasnian Kellwasser biocrises differed markedly from each other in their intensities, as shown by both the number of extinctions and the taxonomic ranks concerned (Fig. 6). In the Lower Kellwasser Event, 13 of the 22 genera present in the mid-Frasnian strata vanished, whereas all eight families crossed into the late Frasnian. Of these, five families and all the remaining ten genera became extinct at the base of the Upper Kellwasser Event. These included *Eskoharpes*, *Gondwanaspis*, and *Telopeltis*, the single last representatives of the orders Harpetida, Odontopleurida, and Corynexochida, respectively. This demonstrates that the Upper Kellwasser Event dramatically affected trilobite biodiversity in higher rank categories but was in fact rather insignificant in the lower ranks, with even fewer genera (10) disappearing at that time than during the period immediately preceding the Lower Kellwasser Event.

Eye reduction occurred in a great number of unrelated lineages that evolved contemporaneously to form the atheloptic assemblages before the terminal Frasnian extinction: *Acuticryphops*, *Trimeroccephaloides* (Phacopidae), *Telopeltis* (Scutelluridae), *Chlupaciparia*, *Pteroparia* (Tropidocoryphidae), *Canningbole*, *Palpebralia* (Proetidae), and *Cyphaspis* (Aulacopleuridae). This phenomenon coincided with and may be a reflection of the global rise in sea level culminating in an appreciable increase in water depth immediately before the Upper Kellwasser Event (e.g. Girard and Feist 1997). At this time, 67% of the known lower rank taxa had reduced eyes or were blind. All these genera and species disappeared in Zone 13b when both the outer platform and peri-reefal environments were perturbed by the major regressive pulse of the Kellwasser Event (Girard et al. 2005; McNamara and Feist 2008).

The Famennian

The second phase of the evolution of trilobite diversity in the Devonian considered in this study differed markedly from the preceding phase. Indeed, generic trilobite diversity stopped declining and even increased, depending on the metrics considered (Fig. 1) – due to a spectacular increase in origination rate to such a level that the still high extinction rate was compensated (Fig. 2a, b). The presence of only two orders (Fig. 3), survivors of the Kellwasser mass extinction, and the complete domination of the trilobite communities by the Proetidae (Fig. 4d) were also characteristic of this phase.

Like the preceding period, the Famennian was punctuated by a series of events whose impacts on trilobite diversity differed in intensity and nature (Walliser 1996; Fig. 6 herein). These events are, in chronological order, as follows: the *Cheiloceras* and Condroz–Enkeberg Events (early Famennian), the *annulata* Event (middle Famennian), and the Dasberg and Hangenberg Events (late Famennian). With the start of the Famennian, the post-event trilobite recovery was slow; no forms have been found in the first conodont zone. Hitherto 11 genera (seven phacopids and four proetids) have been reported from the early Famennian of Europe, North Africa, Southern Urals, Iran, and NW Australia, although no trilobites of that age are yet known from the Americas or eastern Asia. The earliest Famennian trilobite was the blind phacopid *Nephranops*; it occurred in European off-shore sites as early as the Middle *triangularis* Zone. Although it appears in slightly younger strata in the Canning Basin (Upper *triangularis* Zone), the trilobite *Houseops* is evidence that oculated forms were present not long after the Upper Kellwasser Event (Feist et al. 2009). This genus may have derived from an oculated lineage that survived the Kellwasser Events by living in shallow-water reefal to peri-reefal refugia unaffected by hypoxia.

Strikingly, the proetids reappeared later than the phacopids in all outcrops studied. The earliest Famennian proetids appeared within the Middle *crepida* Zone, three conodont zones later than the appearance of the phacopids. The late–early Famennian saw the true onset of the post-Kellwasser rediversification of the trilobites, as well as that of other marine organisms, such as the ammonoids ('*Cheiloceras* Event', Walliser 1985). Two types of communities

were present in photic off-shore environments: blind proetids of cryptogenic origin such as *Formonia*, *Frithjofia*, and *Calybole*, found in basinal facies, and the first cyrtosymbolines, which had large eyes. This realm was extremely common at that time, expanding from Europe/North Africa to the Southern Urals and Eastern Asia on the one hand, and to Iran and NW Australia on the other. Moreover, it can be speculated that the radiation and final predominance of the Cyrtosymbolinae during the late early and middle Famennian times somewhat mirrored the development of uniform biofacies on shallow outer platforms, accompanying convergence of continental margins all around the proto-Tethyan 'gulf'. During this period, several phacopid taxa, both blind and oculated, appeared contemporaneously with the proetids.

After a mid-Famennian interval characterized by lower rates of generic origination, both the phacopids and proetids experienced a second major diversification period in the late Famennian (Dasberg Event; Fig. 6). At that time, about 70% of the genera known from the Famennian were still present. As was the case just before the Kellwasser Event, the overwhelming majority of these trilobites were blind or had reduced eyes and some, such as *Chaunoproetus* and *Helioproetus*, constituted the final stage in the evolutionary trend towards eye reduction (Yuan 1988). Whereas these forms originated and diversified in the European, Kazakhstani, and Eastern Asian deeper off-shore clymenid realm, normally sighted cyrtosymbolines, such as *Pseudowaribole*, the problematic proetine *Perliproetus*, and the phacopids *Omegops*, *Rabienops*, and *Phacops*, were widely distributed in contemporaneous shallow shoal areas in Europe, Southern Urals, and Iran/Afghanistan or on the expanded shallow carbonate platforms in southern Morocco and Central and Eastern Asia. In the latter realm, the reappearance of brachymetopids and proetines after some 30 Ma of pseudo-extinction since mid-Givetian times constitutes a good example of the 'Lazarus' effect. Strikingly, both communities, with the exception of the Americas and Australia, exhibited a high degree of cosmopolitanism.

The terminal Devonian Hangenberg Extinction Event had an extremely severe effect on trilobite evolution, as demonstrated by extermination of the order Phacopida and the high level of generic extinctions, comparable only to those observed during the Middle Devonian events (Fig. 6). The reduced-eyed

and blind taxa among the Proetida became extinct without exception, most likely trapped by the sudden rise in sea level and the spread of oceanic anoxia. The survivors probably originated from conservative, normally sighted communities in shallow shoal refugia that were beyond the devastating effects of the oxygen-depleted conditions at the onset of the Hangenberg black shale. The Hangenberg Event recalls the terminal Frasnian mass extinction in terms of both survivorship and post-event recovery. However, unlike the situation after the Upper Kellwasser Event, post-event recovery was immediate and worldwide after the Hangenberg biocrisis, even in North America where *Pudoproetus* originated in the Late *praesulcata* Zone before the defined D/C boundary (Feist and Petersen 1995). The total worldwide absence of reduced-eyed trilobites in this terminal Devonian period may indicate a considerable eustatic shallowing. The first phillipsiids, representatives of the subfamilies Archegoninae and Weaniinae, developed contemporaneously at that time and crossed the D/C boundary without any loss before their main radiations in the Early Carboniferous (Brauckmann et al. 1993). A major biotic turnover in trilobite diversity was associated with the Hangenberg Event, i.e. shortly before the internationally defined D/C boundary. The subsequent D/C lineages have yet to be defined within the continuously radiating trilobite lineages of 'Carboniferous aspect' (Chlupáč et al. 2000).

The Early Carboniferous

The Carboniferous began with high levels of raw, standing, and to a lesser extent normalized diversities (Fig. 1) resulting from a high origination rate associated with a moderately low extinction rate (Fig. 2a, b). This period was characterized by a great diversity of available habitats, an element that probably played a major role in the emergence and radiation of the Phillipsiidae after the late Famennian Hangenberg Extinction Event. Indeed, facies constraints on the composition of the trilobite communities at that time are evident (Hahn 1990; Owens 1990). For example, the 'Culm facies', representing the bathyal zone, has yielded rich and diversified faunas composed mainly of reduced-eyed or blind representatives of the two most-diversified subfamilies of the Tournaisian: the Archegoniinae and the Cystispirinae (Hahn 1990;

Yuan and Xiang 1998). The other phillipsiid subfamilies present during this stage occupied shallower environments represented by limestone facies (e.g. the 'Carboniferous limestone' of Central Europe). Though the 'Culm facies' developed in Asia (Hahn et al. 1989; Yuan and Xiang 1998; Hahn and Hahn 2005) and especially in Europe, where it has been intensively studied (e.g. Hahn and Hahn 1981), limestone facies dominated the Mississippian of North America (Brezinski 1998) and Australia (Engel and Morris 1997). In these shelf habitats, the palaeogeographic context might have favoured diversification of the trilobites. Indeed, the trilobite faunas of the limestone facies in Europe are mainly composed of phillipsiines, griffithidines, cummingellines, and bollandiines (Phillipsiidae; Hahn 1990). In contrast, the Australian faunas are particularly rich in linguaphillipsiines and weaniines (Phillipsiidae), as well as crassiproetines (*Conophillipsia*, Proetidae; Engel and Morris 1997), and although present worldwide, ditomopygines dominated the trilobite communities in North America.

Trilobite diversity fell during the Viséan (Fig. 1) with a significant reduction in origination rate and a simultaneous increase in extinction rate (Fig. 2a, b). However, the raw and standing diversities were still rather important; the distributions and compositions of trilobite communities remained grossly unchanged. The origination rate approached zero during the Serpukhovian, whereas the extinction rate, although diminishing, remained at around 0.35 (Fig. 2a, b). Accordingly, trilobite diversity decreased strongly to a value similar to those observed during the Late Devonian (Fig. 1a). The continuous fall in trilobite diversity during the Middle/Late Mississippian was accompanied by a continuous reduction in turnover rate (Fig. 2c). In Eastern and Central Europe, as well as in the British Isles, extinction of the diversified 'Culm' faunas resulting from abolition of deeper water basinal environments during the Sudetian orogenic phase might have accounted for a large part of the fall in trilobite diversity during the late Early Carboniferous (Hahn 1990; Owens 1990). Brezinski (1999) also noted a major fall in trilobite diversity in the USA during the middle Viséan (mid-Meramecian). Following a period of exceptional diversity, his stage 2, which lasted until the end-Serpukhovian, was characterized by an extremely limited number of genera. He attributed this significant drop in trilobite diversity to

a transition from a transgressive period to a regressive period (Brezinski 1999). In eastern Australia, local volcanic activity led to the first episode of reduced diversity at the beginning of the Viséan. After moderate diversification, a second episode of diversity decline at the end of the Viséan might have resulted from the cumulative effects of a major local volcanic episode and rapid cooling marking the onset of a period of glaciation in Australia (Dickins 1996; Engel and Morris 1997).

As with the trilobites, the generic diversity of the marine invertebrates as a whole was rather high in the Tournaisian, declining throughout the Early Carboniferous (Alroy et al. 2008; Fig. 1). A long-term drop in sea level apparently paralleled this reduction in diversity of marine invertebrates (Haq and Schutter 2008). The late Mississippian–early Pennsylvanian has long been recognized as a period of significant changes in marine ecosystems. Although varying in intensity and duration, mid-Carboniferous events have been described in ammonoids (Saunders and Ramsbottom 1986), brachiopods (Raymond et al. 1990; Shen et al. 2006), rugose corals (Kossovaya 1996; Wang et al. 2006), reefs (Flügel and Kiessling 2002), and conodonts (Ziegler and Lane 1987; Nemirovskaya and Nigmatganov 1994). These perturbations of the biosphere were concomitant with an increase in the magnitude of glacio-eustatic fluctuations; the late Viséan represented the initial phase of ice accumulation; it was followed by a period of widespread glaciation that lasted until the beginning of the Bashkirian (Dickins 1996; Rygel et al. 2008). During this glacial episode, glacio-eustatic fluctuations as great as 40–100 m might have occurred (Rygel et al. 2008). Important eustatic changes recorded during the late Mississippian were already regarded as a major eustatic event by Saunders and Ramsbottom (1986), with significant repercussions for faunal diversity. However, the assumption of these and other authors (e.g. Ziegler and Lane 1987; Weems 1992) that a major extinction event occurred during the Serpukhovian is not really supported by our data for trilobites. Indeed, even if a marked drop in trilobite generic diversity could be observed during this stage, it was already preceded by a diversity decline during the Viséan (Fig. 1a, b). This is particularly obvious when the durations of the time intervals are considered (Fig. 1c). Moreover, the origination and turnover rates had already declined during the Viséan (Fig. 2a, c). A similar pattern of diversity has been

described for the Carboniferous brachiopods of South China by Shen et al. (2006). These authors concluded that the ‘mid-Carboniferous extinction’ might have been a gradual process, experienced over a long time, and was mainly associated with a restructuring of brachiopod communities. We regard the decline in trilobite generic diversity and extinction of three subfamilies (Cystispiniinae, Linguaphillipsiinae, and Phillipsiinae) during the late Mississippian (late Viséan–Serpukhovian) to have been similarly gradual. Global climatic and eustatic changes (i.e. the glacial episode initiated during the Viséan) are thus regarded as the main factors responsible for evolution of trilobite diversity at that time. However, tectonic evolution (e.g. in Europe) or volcanic activity (e.g. in Australia) might also have had significant effects on trilobite diversity locally.

The Bashkirian–Moscovian Period

The pattern of the evolution of trilobite generic diversity during the Bashkirian and Moscovian is not clear. Whereas the raw diversity remained roughly unchanged, the standing diversity declined moderately and the normalized diversity increased (Fig. 1a–c). However, the origination, turnover, and to a lesser extent the extinction rates increased significantly during the Bashkirian and remained roughly unchanged during the Moscovian; this points to a certain renewal within the trilobite communities. Thus, about 70% of the trilobite genera that occurred in the Kasimovian originated after the Serpukhovian. Considering that the origination rate in the Kasimovian was extremely low, this suggests that most of this renewal occurred during the 11 million years or so of the Bashkirian–Moscovian interval, a period only slightly exceeding the Serpukhovian stage (Ogg et al. 2008). This renewal of trilobite communities was accompanied by continued restructuring initiated during the late Early Carboniferous. Ditomopygines become particularly common, whereas archegoniines became rare (Fig. 5) until their simultaneous extinction with another phillipsiid subfamily, the Griffithidinae, and the family Aulacopleuridae during the Moscovian. This second phase in evolution of the diversity and composition of post-Devonian trilobites ended with a reduction in generic diversity and falls in the origination, extinction, and turnover rates during the Kasimovian (Figs. 1 and 2).

The diversified faunas of Spain (Gandl 1987, 2011) and China (Hahn et al. 1989; Yuan and Li 2000) illustrate particularly well the transitional aspects of the early Late Carboniferous. These faunas were dominated by cummingelline and ditomopygine phillipsiids and brachymetopids, but still included a notable proportion of archegoniine trilobites. However, in the USA, the early Late Carboniferous was associated with a renewal within the trilobite communities, rather than a modification of their subfamilial compositions. The prevalence of ditomopygines was already apparent in the late Mississippian trilobite communities composed of species of *Kaskia* and *Paladin* (Brezinski 2003, 2008). These trilobites were replaced during the late Bashkirian by another moderately diversified fauna composed exclusively of ditomopygines (*Sevillia*, *Ditomopyge*, and *Ameura*; Brezinski 1999). Although poorly known (Hahn and Hahn 1991, 1992), the late Early Carboniferous trilobites of Alaska were more diversified than those of the USA, comprising representatives of four phillipsiid subfamilies. They were replaced in the early Late Carboniferous by a fauna composed of brachymetopids and once again, various phillipsiids (Hahn and Hahn 1991, 1992, 1993). The Ditomopyginae only slightly dominated this trilobite association in terms of generic diversity, but the genus *Ditomopyge* was by far its most abundant component (Hahn and Hahn 1991). Therefore, the Bashkirian–Moscovian period in Alaska was associated with both a renewal of trilobite communities and the onset of the ditomopygine predominance, at least in terms of abundance. In eastern Australia, the Late Carboniferous trilobites were only represented by a single species of the brachymetopid *Australosutura* (Engel and Morris 1997). This species belonged to the presumably cold-water *Levipustula* fauna (Campbell 1961; see Dickins 1996 for a different interpretation). A similar fauna has been described in the Bashkirian of Argentina; it also included a representative of *Australosutura* (Hahn et al. 2001). However, diverse species of the ditomopygine *Ameura* have also been found in the Late Carboniferous of South America. These trilobites, known from different regions of Brazil, are apparently of Moscovian age (Hahn et al. 2001); this suggests that trilobites were also mainly represented by ditomopygines in this part of South America at that time. Lastly, two early Late Carboniferous trilobite faunas showed unusual compositions with regard to the Ditomopyginae. Weber

(1937) has described Bashkirian and Moscovian trilobite faunas from western Russia (Urals, Moscow Basin) and the Ukraine (Donetz Basin) composed exclusively of ditomopygines. This unique situation might be due to these faunas having come from shale facies (Hahn et al. 1989). In contrast, contemporaneous faunas in Japan were completely devoid of ditomopygine trilobites (Kobayashi and Hamada 1980) but were especially rich in cummingellines. They therefore anticipate the importance this group would have in later periods.

A rather modest increase in diversity of marine invertebrates apparently occurred in the early Late Carboniferous (Alroy et al. 2008; Fig. 1). This increase was particularly well exemplified by brachiopods from South China (Shen et al. 2006); they displayed a notable increase in origination and turnover rates indicative of a radiation. Rugose corals from the same region also diversified during this period and as with the trilobites, they illustrate a profound restructuring of benthic communities with replacement of large dissepimented solitary corals by compound corals (Wang et al. 2006). According to Wang et al. (2006), these compositional changes in marine benthic communities were associated with a transgression that led to formation of a uniform carbonate platform in most areas of South China. Major transgressive events were also associated with the Bashkirian and Moscovian in the Tethys Himalaya (India, Nepal, and Tibet) and in the peri-Gondwanan belt (Garzanti et al. 1998). On the global scale, the Bashkirian and most of the Moscovian have been considered an interval of low-magnitude eustatic sea-level changes indicating a reduction in the worldwide volume of glacial ice (Rygel et al. 2008; see also Isbell et al. 2003). This period also marked the onset of a long-term rise in sea level lasting until the beginning of the Permian (Haq and Schutter 2008). In summary, restructuring of the trilobite communities and, more generally, the benthic marine communities that occurred during the early Late Carboniferous might have resulted from favourable eustatic and climatic conditions.

The Kasimovian–Roadian Period

This long interval (about 40% of the Late Palaeozoic), which encompassed the late Late Carboniferous, the Cisuralian, and the earliest Guadalupian, was

characterized by invariably low raw and standing generic diversities (Fig. 1) and low origination, extinction, and turnover rates (Fig. 2). The only notable changes during this period were two modest peaks in the turnover rate corresponding to an increase in the extinction rates of the Asselian and an increase in the origination rates of the Kungurian. In terms of composition, cummingellines and weaniines seem to have been relatively more diversified during the Kasimovian–Sakmarian and Sakmarian–Roadian intervals, respectively (Fig. 5), but the raw diversity was too low to allow any reliable estimation of the relative diversities of the phillipsiid subfamilies. Basically, the trilobite communities remained roughly unchanged during most of this period. For example, more than 80% of the genera present in the Asselian originated or already existed in the Moscovian. The extinction of four of those genera by the end of the Asselian was not accompanied by an equivalent origination, and therefore led to a slight drop in diversity. It was only during the Kungurian that definite renewal began, but it remained modest until the Wordian.

This third phase in the evolution of trilobite diversity in the Late Palaeozoic, characterized by low diversity, few changes, and a predominance of ditomopygines, is particularly well illustrated in the USA. After the extinction of *Sevillia* during the Moscovian, only two ditomopygine genera (*Ditomopyge* and *Ameura*) occurred during most of the Late Carboniferous (Brezinski 1999). More diversified trilobite faunas were present during the terminal Carboniferous and at the beginning of the Permian. However, these faunas did not result from local diversification but testify to an invasion of the American biota by pandemic trilobites (Brezinski 1999). It is also noteworthy that the last diversification phase in the history of the trilobites had begun by the end of the Cisuralian in the USA (see stage 4 of Brezinski 1999), whereas it became obvious no sooner than the Wordian on the global scale. Recently, Owens (2003) noted that all genera that occurred in the Tethyan Realm during the early Cisuralian were inherited from either the Early Carboniferous or the Late Carboniferous. Our data suggest that the genera that originated during the Bashkirian or later actually represented the overwhelming majority of trilobites on the global scale. Interestingly, some trilobites occurred at high latitudes after the Asselian. In particular, the reappearance of trilobites in south-eastern Australia (Tasmania) in

the late Sakmarian (Wass and Banks 1971) might indicate a certain amelioration of the climatic conditions locally. However, their persistence during the Artinskian, together with the appearance of *Anujaspis* in high northern latitudes (Balashova 1960) might indicate somewhat warmer climatic conditions on a more global scale.

The end-Carboniferous and early Cisuralian times (Gzhelian to early Sakmarian) are recognized as the largest and most widespread period of Late Palaeozoic continental glaciation (Isbell et al. 2003; Montañez et al. 2007; Rygel et al. 2008; Tabor and Poulsen 2008). According to Alroy et al. (2008), this period was associated with a particularly low diversity of marine invertebrates; its end might have allowed a considerable increase in their diversity, culminating during the Guadalupian. The late Sakmarian to Artinskian stages thus represent a transitional interval associated with compositional changes within marine benthic communities (e.g. fusulinids: Leven et al. 1996; reefs: Weidlich 2002; rugose corals: Kossovaya et al. 2001; also see Clapham and James 2008). Whereas a slight fall in the normalized trilobite diversity (Fig. 1c) and a modest increase in the extinction and turnover rates (Fig. 2b, c) can be tentatively associated with the climax of the Asselian glaciation, no significant changes in the trilobite diversity parameters really indicate the onset of an icehouse/greenhouse climatic transition in the mid-Cisuralian. As mentioned above, this latter transition is nevertheless suggested by the geographical distribution of the trilobites, especially the reappearance of taxa at high latitudes. The modest increases in the origination and turnover rates (Fig. 2a, c) observed in the Kungurian might indicate the onset of the last diversification phase of trilobite history. However, the raw diversity had reached such low values after the Kasimovian that any interpretation of the evolution of trilobite diversity during this period must be made with great caution.

The Wordian–Changhsingian Period

In contrast to the preceding phase, the last stage in the evolution of trilobite diversity was associated with notable changes. As indicated by the rise in the raw and normalized generic diversities (Fig. 1a, c) and the simultaneous increase in origination rates (Fig. 2a), the Wordian represented a period of trilobite

diversification. The renewal within trilobite communities is exemplified by the fact that 45% of the genera present during this stage were new. However, as mentioned above, this renewal occurred earlier in the USA (i.e. during the Kungurian; Brezinski 1992, 1999) with the appearance of an endemic fauna exclusively composed of ditomopygines (Lerosey-Aubril 2008). It is also noteworthy that most of the new genera belonged to the Ditomopyginae. Although modest, this diversification of trilobites in the Wordian mirrored a general increasing trend in the diversity of marine invertebrates, initiated in the mid-Cisuralian (Alroy et al. 2008).

More than 50% of the genera that originated during the Wordian were restricted to this time interval. This observation stresses another characteristic of this stage: the rather high extinction rates (Fig. 2b). These remained high in the Capitanian and the Wuchiapingian, whereas origination rates decreased strongly during these two stages (Fig. 2a). The end-Guadalupian was thus characterized by an important reduction in the generic diversity of trilobites (Fig. 1; Owens 2003) and therefore deserves particular attention. A major drop in diversity has also been recorded in brachiopods (Shen and Shi 2002), corals (Wang and Sugiyama 2000), gastropods (Erwin and Pan 1996; Pan 2004), fusulinids (Stanley and Yang 1994; Ota and Isozaki 2006), reefs (Weidlich 2002), and other taxonomic groups (e.g. Yin et al. 2007) which has led several authors to propose a mass extinction in the late Guadalupian (e.g. Stanley and Yang 1994; Shen and Shi 2002; Ota and Isozaki 2006; Retallack et al. 2006; Ali and Wignall 2007; Weidlich and Bernecker 2007). Whether it should be recognized as an independent extinction event is still contentious (Clapham et al. 2009), but most authors now admit that the end of the Guadalupian marks the onset of a period of major environmental perturbation. Global cooling ('Kamur Event'; Isozaki et al. 2007a, b; Clapham et al. 2009), a major regression (Erwin 1993; Shi and Shen 2000; Shen and Shi 2002; Tong 2004; Pan 2004; Yin et al. 2007), a massive spread of deep-water anoxia (Bottjer et al. 2008; Clapham et al. 2009), a catastrophic methane release (Retallack et al. 2006), and an eruption of a large volume of basalts ('Emeishan Large Igneous Province'; Ali et al. 2002; Ali and Wignall 2007) are all environmental changes identified/suspected in the late Guadalupian that might have ultimately resulted in a reduction in

biodiversity. Considering the low diversity of trilobites in the Permian, it might be unwise to use their diversity patterns to favour one or other of these hypotheses. However, our data show that the origination rate decreased strongly during the Capitanian and Wuchiapingian, whereas there was no increase in the extinction rate (Fig. 2a, b). Clapham et al. (2009) have recently described a similar pattern in several other groups of marine invertebrates; they interpret this as evidence that end-Guadalupian diversity losses resulted from a lack of originations rather than from increased extinctions. According to them, this low level of origination in the Capitanian–Wuchiapingian can only be explained by (1) cooling, (2) the spread of anoxic deep-water masses, or (3) a reduction in area and provinciality of the marine habitat – attributable to tectonic activity.

The Changhsingian was also marked by an absence of originations: the trilobites were then represented by only four genera – *Acropyge* (Ditomopyginae), *Carniphillipsia* (Ditomopyginae), *Kathwaia* (Bollandiinae), and *Paraphillipsia* (Cummingellinae) – to which might be added *Pseudophillipsia* (Ditomopyginae) and *Cheiropyge* (Brachymetopidae), known as several species of 'Lopingian' age. Although the trilobites persisted in several peri-Tethyan regions (Lerosey-Aubril 2008), they were extremely diminished in term of their diversity and were apparently unable to accommodate the dramatic environmental upheavals of the end-Permian. Therefore, they became extinct after 270 million years of a remarkable evolutionary history.

Conclusion

The time period considered in this study encompasses more than 145 My (Ogg et al. 2008) and is therefore particularly rich in biotic, climatic, eustatic, geochemical, and tectonic events. For example, the dynamic convergence of Lower Palaeozoic continental blocks, ending with formation of the mega-continent Pangaea during the Permian, certainly constituted a singular tectonic context (Scotese and McKerrow 1990). Dramatic climatic changes also occurred in this period, including the transition from a greenhouse to an icehouse climatic system around the Early–Late Carboniferous boundary and vice versa during the Cisuralian. This multiplicity of major

environmental changes makes it difficult to assess the precise impact of each of these events on the biosphere or to discern how they interacted. Consequently, it is not possible to determine with certainty which of these changes ultimately influenced trilobite diversity.

However, in the Middle and Late Devonian, the stratigraphic precision of the quantitative data allows us to recognize that eustatic variations had a major influence on trilobite diversity, even if the origins of these eustatic changes are not always clear. The expansion of outer-shelf domains during transgressions or the increases in near-shore facies belts during sea-level low stands apparently had important repercussions on biotic diversity, determining the available benthic habitats successfully invaded by biotas during adaptive radiations. The trilobites were particularly sensitive to bathymetric changes, as demonstrated by the generalized trend to eye reduction within trilobite communities during major transgressive phases (e.g. Frasnian, Famennian) or the very many extinctions associated with abrupt regressive events (e.g. Upper Kellwasser Event, Hangenberg Event). Therefore, it seems reasonable to infer that the intensity and dynamics of eustatic oscillations had direct consequences for trilobite evolutionary history at that time. Deepening trends led to important morphologic and ecologic adaptations of different taxa to life in deep-benthic habitats; brutal regressive pulses triggered massive extinctions within these specialized communities. From the Eifelian to the Frasnian, a series of extinction events led to drastic impoverishment of trilobite diversity, involving in the first place, and nearly exclusively, the lower taxonomic categories (genus and species). It was only at the very end of this period, when trilobite generic diversity had dropped to the lowest level of Early to Mid Palaeozoic times, that higher ranks were severely affected – at the end-Frasnian biocrisis. In the Famennian, when the group was represented by only two orders, originations started to compensate for the still high extinction rate, marking the onset of remarkable lower rank diversification. Although this second phase ended with the major turnover caused by the terminal Famennian Hangenberg Event, the general diversification trend continued and even accelerated in the Tournaisian when extinction rates became more moderate.

The Late-Palaeozoic diversity patterns described here suggest there may have been a relationship between the evolution of trilobite diversity and the main climatic changes in post-Devonian times. When glacial conditions occurred, whether they were associated with individual events (Viséan and Serphukovian glaciations, Capitanian Kamura Event) or with an icehouse period (Kasimovian–Sakmarian), the origination rate fell dramatically (Fig. 2a), leading to a drop in diversity or its maintenance at extremely low levels (Fig. 1). However, the stratigraphic resolution of our recently compiled database is not sufficient to explore this relationship further, because it does not allow identification of rapid fluctuations in diversity or their correlation with abrupt climatic changes. In this regard, it would be interesting to consider time intervals shorter than stages in the Early Carboniferous because these are notably longer than those of the Late Carboniferous and Permian. This should permit better appreciation of the dynamics of the diversity fluctuations during this crucial period in the late evolutionary history of the trilobites. Another potentially important improvement to the database would be incorporation of palaeobiogeographic data; this would allow us to assess the completeness of the fossil record for each interval and to search for palaeobiogeographic signals of major events, especially climatic events. It would also be worth including data on ecologic diversity from both morphologic and taphonomic studies because they might allow us to describe the evolution of the roles of trilobites within Late Palaeozoic ecosystems.

Finally, this quantitative study has pointed out a particularly interesting aspect of the late evolutionary history of the trilobites. After the terminal Frasnian Kellwasser Event, there was an obvious decoupling of trilobite diversity at the generic level and trilobite diversity at the familial and ordinal levels. The spectacular increase in generic diversity observed in the Tournaisian, with values not attained since the Eifelian, demonstrates that trilobites still had an intrinsic and significant capacity to adapt at that time. This increased diversity involved only the Proetida, a conservative group that had hitherto maintained its very simple morphological bauplan since Cambrian times. It then underwent a burst of homeomorphic

variation, adapting to various ecological behaviours and habitats formerly occupied by morphologically different groups, such as the Phacopida. Therefore, extinction of this class of emblematic arthropods at the end of the Permian was not, as frequently proposed, an almost inevitable fate after restriction of the group to one single order after the Late Devonian biocrises. Instead, it more probably represents the ultimate result of major environmental changes that affected the biosphere during the Late Palaeozoic. Much more attention must be paid to Permo-Carboniferous trilobites if we are to fully understand this extinction.

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